

Fabrication of an anti-fouling coating based on epoxy resin with a double antibacterial effect via an in situ polymerization strategy

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ABSTRACT

The hull and subsea equipment are often plagued by widespread marine biofouling, so developing a sustainable environmental protection antifouling paint is urgent. The study designed and developed an antifouling composite coating system (E44-Ag-PGI) with a synergistic antibacterial effect based on the dual strategy of the bacteriostatic effect of stereochemical structure on polymer surface and release sterilization of metal nanoparticles. Firstly, an antifouling agent (PGI) was prepared by simple radical polymerization by combining glycidyl methacrylate with borneol, a complex bicyclic monoterpene structure extracted from plants. Then bisphenol A epoxy resin with strong adhesion to metal was selected as the base material. The synthetic antifouling agent with epoxy groups was polymerized in situ into the epoxy resin network, and compared with pure epoxy resin, the elongation at break of E44-Ag-PGI coating with 20 % PGI content increased from 40.37 % to 124.8 %, which was mainly attributed to the introduction of a soft chain of antifouling agent polymer, which improved the flexibility of the coating. Meanwhile, the antibacterial efficiency of the coating against *Pseudomonas aeruginosa* and *Staphylococcus aureus* was 97.69 % and 95.9 %, respectively. This excellent antibacterial performance was mainly attributed to the synergistic antibacterial action of natural borneol and AgNPs. In addition, the biofilm and diatom resistance experiments of the coating showed that the coating also has an excellent antifouling effect. This provided a new research idea for environmentally friendly antifouling coatings.

1. Introduction

Marine biofouling is a phenomenon caused by the settlement and reproduction of fouling organisms on the surface of underwater equipment [1–4], and this undesirable attachment is a complex interactive process [5,6]. This phenomenon affects the average service time of marine equipment. It seriously involves economic development, environmental protection, exploitation of marine resources, and human health, which has undoubtedly become a global problem [7]. In response to this problem, researchers have developed and designed numerous protection methods, such as the method of mechanical removal, ultrasonic cleaning, electrochemical, and the application of antifouling coatings, among which the application of marine antifouling coatings is considered to be one of the most effective methods to solve marine biofouling [8]. Traditional antifouling coatings, such as tributyltin, have been explicitly banned by the International Maritime

Organization (IMO) in 2008. Since their high teratogenicity and severe hazards to the marine environment, the subsequent development of copper-based antifouling coatings has been limited by the high copper content in ports and offshore [9–13]. Hence, designing and developing a non-toxic or low-toxic, environmentally friendly antifouling coating is urgent.

Polydimethylsiloxane (PDMS) has become one of the vital role players in marine antifouling coatings due to its low surface energy, low modulus, and low adhesion to hard dirt [14–16]. However, the weak adhesion of silicone coatings to substrates and their tendency to peel off make the antifouling strategy fail to achieve the desired effect [17], thus limiting their use to some extent. Although many methods to improve the adhesion of silicone-based coatings have also been developed in recent years, such as introducing polar interactions [7], adding inorganic particles [18,19], or introducing urea groups [17], these methods still suffer from high cost, poor adhesion, cumbersome processes and

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most importantly, they can destroy the original characteristics of silicone-based coatings [7]. In addition, epoxy resins are also widely used as coating substrates. It has excellent adhesion, cohesiveness, and stability. This establishes a good foundation for its application in marine anti-corrosion and antifouling coatings [20,21]. Among them, bisphenol A-based epoxy resin (EP) coatings with excellent scratch hardness, adhesion properties, and other characteristics have a broad application prospect as coating substrates. However, EP coatings are not inherently antifouling, so they also quickly lose their inherent corrosion resistance in harsh environments. Therefore, biocides are often introduced into the epoxy resin matrix to prevent bacterial infection and biological adhesion [22–25].

In recent years, researchers have made many efforts to develop biocides to address the problem of bacterial adhesion to coatings and subsequent proliferation. Initially, organotins were banned as biocides because they severely disrupted the ecological balance. Then came inorganic biocides, mainly silver, zinc, copper, and titanium dioxide [26]. The development and design of natural biological antifouling agents have become a hot research topic in recent years. Terpenoids, camptothecins, and phenolic compounds from natural resources are reported to have the ability to resist the adhesion of marine biofouling [27–30]. Among them, borneols, a natural compound extracted from many herbal plants with a unique crystalline cyclic alcohol structure, are one of terpenoids. The antibacterial mechanism of borneols is mainly attributed to its surface stereochemical structure that affects the surface recognition of bacteria, and this unique structure could regulate the interaction between microorganisms and the material surface to achieve the effect of preventing bacterial adhesion, thus giving it a place in the field of anti-fouling and antibacterial. In recent years, borneols and their derivative materials have been studied. The formation of polymers will greatly enhance surface characteristics or functions [31–35]. Cheng et al. combined zwitterion (MPC) with borneol (BA) by free radical polymerization and prepared surfaces with excellent antifouling and antibacterial properties by Schiff base bonding [36]; Yang et al. also developed and designed a water-soluble antibacterial agent based on borneol with multiple drug resistance [35]. However, if borneol or its derivatives are added directly to the substrate, the coatings will face the problem of uncontrolled release, which reduces the antifouling performance of the coating. Therefore, finding suitable methods to introduce borneol-based compounds into the resin is a new challenge [37–39].

Hence, in this work, based on the dual strategies of polymer surface stereochemistry and release sterilization, combined with the advantages of natural borneol and AgNPs, a new antifouling coating of epoxy resin with synergistic antibacterial action was designed. In addition, we studied the coatings of mechanical properties, thermal stability, surface morphology, wettability, and resistance to marine bacteria, biofilms, and diatoms. The results showed that the E44-Ag-PGI coating had strong antifouling ability. This study provided a promising strategy and more possibilities for developing marine antifouling coatings.

2. Experimental section

2.1. Materials

Glycidyl methacrylate (GMA); Isobornyl methacrylate (IBOMA); amino-terminated polyether amine (D400), and 2-hydroxy-1-ethanethiol were purchased from Aladdin Chemical Reagent Co., Ltd. (China) and used as received without further purification. Epoxy resin (E44) and 2,2'-Azobis(2-methyl propionitrile) (AIBN) were purchased from Shanghai Macklin Biochemical Co., Ltd. Nano Silver (AgNPs) was purchased from Pioneer Nanomaterials Technology Co. LTD. (China). Tetrahydrofuran (THF) and methanol were purchased from Tianjin Fuyu Fine Chemical Co., Ltd.

2.2. Synthesis of PGI

An environmentally friendly antifouling agent was synthesized by free radical polymerization, as indicated by Fig. 1a, using AIBN as an initiator and controlling the molecular weight of the polymer. In a typical experiment, GMA (2.84 g, 20 mmol), IBOMA (17.79 g, 80 mmol), β -mercaptoethanol (0.78 g, 10 mmol), and AIBN (0.17 g, 0.8 wt%) were dissolved with THF and added to a three-neck flask with a reflux condensing device. Oxygen was removed by bubbling with N_2 for 30 min, and the polymerization process was stirred in an oil bath at 65 °C for 11 h.

After the reaction, the product was cooled to room temperature, precipitated with excess methanol, and dried under vacuum at 70 °C for 24 h to obtain a white product. The resulting polymerization product was named PGI. The molecular weight of the polymer was characterized by gel permeation chromatography (GPC), where the polydispersity index (DM) and Mn of PGI were 1.532 and 3040 g/mol, respectively.

2.3. Preparation of coatings

All substrates (glass sheets, tinplate sheets, and Q235 mild steel sheets) were treated with ethanol and deionized water in an ultrasonic bath for 30 min and then dried with nitrogen gas. The polymer powders (PGI) and AgNPs were dissolved and dispersed into THF, mixed with E44, ultrasonically dispersed for 30 min, and then added to D400, which were evenly mixed and coated on the substrate by drip. Finally, the coating was procured at room temperature for 6 h and then cured in an oven at 55 °C for another 10 h to obtain the final coating. (The dry film thickness measured by the paint thickness gauge QNIX4500 was approximately 300 μ m). The curing mechanism and detailed coating formulation were also shown in Fig. 1b and Table S1.

2.4. Characterization

Fourier Transform Infrared Spectroscopy (FTIR). FTIR spectra of polymers in the range 4000–400 cm^{-1} were measured by FTIR spectrometer (Nicolet IS10, heat-Nicolet). Proton nuclear magnetic resonance spectroscopy (1H NMR). The proton NMR spectra of the polymers were measured by a Bruker spectrometer model Avance NEO 400 MHz, and $CDCl_3$ was selected as the solvent. X-ray photoelectron spectroscopy (XPS). X-ray photon spectra of E44-Ag-PGI-20 were collected in an AXIS SUPRA X-ray photoelectron spectrometer (Kratos, UK). Gel permeation chromatography (GPC). The molecular weight (Mn) and polydispersity (DM) of PGI were characterized by measuring the GPC of polymer PGI on a PL-GPC50 instrument. Thermogravimetric analysis (TGA). Thermogravimetric analysis was performed using TG209F1 (NETZSCH) under a nitrogen atmosphere. Samples were taken about 5 mg at a time and scanned for each sample at a heating rate of 15 °C/min between 30 °C and 600 °C. Scanning electron microscopy (SEM). Scanning electron microscopy images and Energy dispersive X-ray spectroscopy elemental maps (EDS) were obtained on a JSM-6700F instrument. Atomic force microscopy (AFM). The surface profile of the coating before and after immersion was tested by Bruker Dimension Icon instrument. Differential Scanning Calorimetry (DSC). Differential scanning calorimetry was used for characterization tests on a DSC204F1 device (Netzsch, Germany). The scanning range was from –50 °C to 150 °C, and the scanning rate was 20 °C·min $^{-1}$. Dynamic Mechanical Analysis (DMA). The dynamic mechanical analysis test was characterized by the TA Q800 instrument. The loss factor ($\tan \delta$) and storage modulus (E') were monitored at a heating rate of 3 °C·min $^{-1}$ in the temperature range of 0 to 150 °C. DMA measurements were also used to evaluate the modulus in the rubbery plateau and to quantify the cross-linking density ν (moles of crosslinking sites per unit volume, mol·cm $^{-3}$) of each sample based on the rubber elasticity theory (eq. 1):

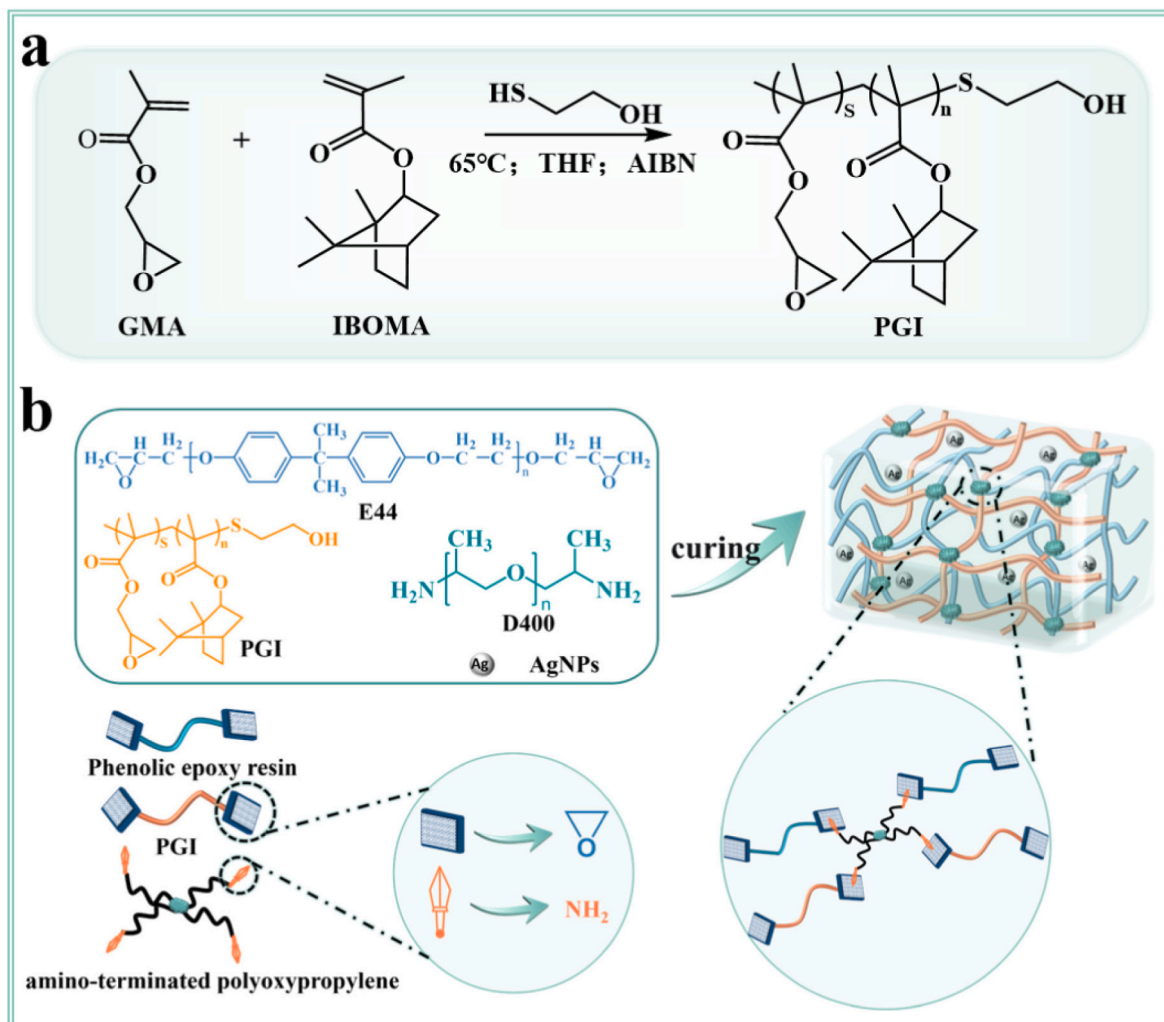


Fig. 1. (a) Synthetic routes of copolymer PGI. (b) Curing diagram of epoxy antifouling coating.

$$\nu = \frac{G'_R}{R \cdot T_C} \quad (1)$$

where T_C is the characteristic temperature and G'_R is the shear storage modulus in the rubbery plateau at T_C . G'_R is correlated to the tensile storage modulus in the rubbery plateau E'_R based on eq. 2, in which n is the Poisson's ratio (equal to 0.5 for elastically deformed incompressible isotropic materials). T_C is taken at 50 °C above the $T_{g,DSC}$ of the sample under test.

$$G'_R = \frac{E'_R}{2 \cdot (1 + n)} \quad (2)$$

2.5. Tensile tests

A tensile test was performed by CMT4204 universal testing machine. The sample was made into a dumbbell type (length × width: 50 × 4 mm³) and tested at room temperature at a 50 mm/min stretching rate.

2.6. Shore hardness tests

The test of coating hardness was performed by Shore D hardness tester GS-702 N (TYPE D), which was executed according to the national standard GB2411-78-1980. The sample was prepared as a film of 6–7

mm thickness and measured 10 times to take the average value.

2.7. Adhesion tests

The adhesion of the coating was evaluated according to ASTM D4541-09 and ASTM D 3359-09.

2.8. Contact angle and surface energy tests

The water and oil contact angle (WCA) of the coating was measured by the CA-200 contact angle test instrument. The test operation is as follows: 2 μL of deionized water or 1 μL diiodomethane was dropped onto the surface of the coating at room temperature using the fixation method. Each sample was measured five times separately and averaged to obtain an accurate number of contact angles. Finally, the Owens two-liquid method was used to calculate the free energy of the coating surface [40,41]. The formula is as follows:

$$\begin{aligned} \sigma_s^p &= [(137.5 + 256.1 \times \cos\theta_{H_2O} - 118.6 \times \cos\theta_{CH_2I_2})/44.92]^2 \\ \sigma_s^d &= [(139.9 + 181.4 \times \cos\theta_{CH_2I_2} - 41.5 \times \cos\theta_{H_2O})/44.92]^2 \\ \sigma_s &= \sigma_s^p + \sigma_s^d \end{aligned} \quad (3)$$

where, $\cos\theta_{H_2O}$ and $\cos\theta_{CH_2I_2}$ are the radian values of the contact angle of the deionized water and the diiodomethane of the coating, respectively. σ_s^p is the polar force in the surface energy, σ_s^d is the dispersion force in the

surface energy and σ_s is the surface energy of the sample. The unit of surface free energy is mJ/m^2 .

2.9. Silver ion stripping performance tests

The concentration of 0 mg/L, 0.05 mg/L, 0.1 mg/L, 0.25 mg/L, 0.5 mg/L, and 1.0 mg/L was configured with 1000 mg/L of silver standard solution, and the absorbance curve was plotted against the standard solution, and the E44-Ag-PGI-20 coating was immersed in deionized water. The concentration of silver ions was measured by UV-8000 UV spectrophotometer.

2.10. Antibacterial tests

Pseudomonas aeruginosa (*P. aeruginosa*) and *Staphylococcus aureus* (*S. aureus*) were selected as a typical bacterium to evaluate the antibacterial properties of the coating. First, the proportion of solution made in PTFE mold into the corresponding film. All samples were immersed in PBS buffer solution for 24 h and sterilized by UV for 30 min. Then the samples were mounted in Petri dishes, and *P. aeruginosa* with OD 0.72 (equivalent to 1×10^8) was incubated with the samples at 37 °C for 24 h. The number of adherent colonies was determined by the plate counting method for different ratios of coatings. The number of colonies grown on the solid medium was N_1 . The same operation was performed for the samples used as blank controls, and the number of colonies was recorded as N_2 .

The antibacterial rate (A) of E44-Ag-PGI-x was calculated by eq. (4):

$$A = \left(1 - \frac{N_1}{N_2}\right) \times 100\% \quad (4)$$

2.11. Anti-biofilm tests

The biofilm formation was promoted by extending the incubation time of *Pseudomonas aeruginosa*. 1 mL of bacterial suspension (10^8 CFU/mL) was introduced into a 24-well plate containing samples. After incubating the bacteria at 30 °C for 72 h, the bacterial solution was removed with a pipette and carefully rinsed three times with ASW to remove unattached biofilm. The biofilm was then fixed on the coating with methanol, stained with crystalline violet, and excess crystalline violet was rinsed with ASW, and the biofilm was photographed with a digital camera. Finally, 33 % AcOH was added to dissolve the fuel on the biofilm, and then the absorbance at 589 nm was measured using a microplate reader (Tecan Infinite M200 Pro). The survival rate of the treated biofilm relative to the control was calculated.

2.12. Anti-diatom tests

The seawater was sterilized, and the sterilized seawater and diatoms (*Pleurosigma* sp.) were added to the triangular glass flask in a ratio of 4:1. The nutrient solution was added to the flask in a ratio of 1:1:1. The flask was sealed with sterilized kraft paper and twine. Finally, the diatoms were incubated in a suitable environment at 25 °C for 72 h at a light-to-dark ratio of 14 h:10 h. The flasks were shaken every morning, noon, and night to promote the growth of algal species. After incubation, the diatom suspension was diluted to 1×10^5 cell/mm² with sterilized seawater, and the samples were placed in 24-well plates with the diluted diatom suspension and incubated for 24 h. After incubation, the bottles were carefully rinsed with sterilized seawater to remove unattached diatoms, observed by stereo microscope, and counted in five fields of view. Three parallel samples were used for each coating to obtain the average value. The algal coverage on the coatings was determined by analyzing the stereo microscopy images using ImageJ software [17,42,43].

2.13. Electrochemical impedance spectroscopy tests

Electrochemical impedance spectroscopy (EIS) measurements were recorded with the electrochemical workstation (Chenhua, Chi660E). The calomel reference electrode and platinum counter electrode were mounted in a 3.5 wt% NaCl corrosion test cell, and a coated sample with an exposed area of 1 cm^2 was used as the working electrode. The samples were immersed in 3.5 wt% NaCl solution at 25 °C. The EIS measurements were carried out in the 10 MHz to 0.01 Hz frequency range with a noise voltage of 10 mV. The measurements were repeated three times for different samples.

3. Results and discussion

3.1. Structural analysis

The composition and chemical structure of the PGI copolymer was characterized by FTIR and ¹H NMR. As indicated by Fig. 2a, compared with the infrared spectra of IBOMA, GMA, and 2-Hydroxy-1-ethanethiol, the stretching vibration peak of C=C at 1640 cm^{-1} and the characteristic peak of -SH in the range of $2500\text{--}2600 \text{ cm}^{-1}$ disappeared. At the same time, the peaks at 3450 cm^{-1} , 1725 cm^{-1} , and 914 cm^{-1} were attributed to the characteristic peaks of -OH, C=O, and -CH(O)CH- of PGI, respectively, indicating that the polymer PGI had been synthesized. As shown in Fig. 2b, the absorption peaks at 5.63 and 6.01 ppm belonging to the hydrogen atoms directly connected to C=C were no longer present, and the new chemical shifts at 4.46 and 1.93 ppm correspond to the -CH and -CH₂- of the isobornyl in IBOMA, respectively. And j (4.38 ppm), l (3.32 ppm), and m (2.72 ppm) were the absorption peaks of the GMA component of the PGI, which further proved the synthesis of the polymer PGI. Fig. 2c showed the XPS spectra of the E44-Ag-PGI-20 coating. The XPS analysis confirmed the presence of silver, sulfur, and other elements. Fig. 2d, e, and f showed the XPS spectra of Ag 3d, S 2p, and N 1s on the surface of the E44-Ag-PGI-20 coating, respectively. Among them, two representative characteristic peaks of silver appeared at 367.5 eV and 373.5 eV, respectively [44]. At the same time, the characteristic peaks of S2p at 161.35 eV and N1s at 397.85 eV further indicated that the antifouling agent PGI was in situ polymerized into the three-dimensional network of epoxy resin. Fig. S2 showed the infrared spectra of E44 and E44-Ag-PGI-20 coatings. The disappearance of the epoxy group characteristic peak at 914 cm^{-1} indicated that the coating was completely cured.

3.2. Mechanical properties

The excellent mechanical properties could effectively avoid some physical damage when the coating is used. Therefore, the tensile test of the free film peeled off from the coating was carried out. The stress-strain curve of the coating was depicted in Fig. 3a, where the tensile strength of E44 was 40.98 MPa, and the elongation at break was 22 %. After the introduction of PGI, although the tensile strength decreased to 17.47 MPa, the elongation at break increased to 131 %, indicating that based on the original cross-linking network, the introduction of PGI polymerized soft chains would bind some original cross-linking sites to a certain extent, reduce the cross-linking density of the system. Still, it did not hinder the strong cohesion in the three-dimensional network. PGI polymerized soft chain gave the system better flexibility. (Fig. 3c (2)). It was well known that inorganic nano-filler played a reinforcing role in the polymer field. In this study, AgNPs were introduced based on E44-PGI-x, and the results showed that the tensile strength of E44-Ag-PGI-x was higher than that of E44-PGI-20, which was because the fine structure of AgNPs optimizes the physical entangling behavior between polymer chains. Its surface activity enhanced the interaction between AgNPs and polymer segments, which played an excellent reinforcing role in this polymer system (Fig. 3c (3)). Among them, the tensile strength of E44-Ag-PGI-20 reached 26.77 MPa, and the elongation at

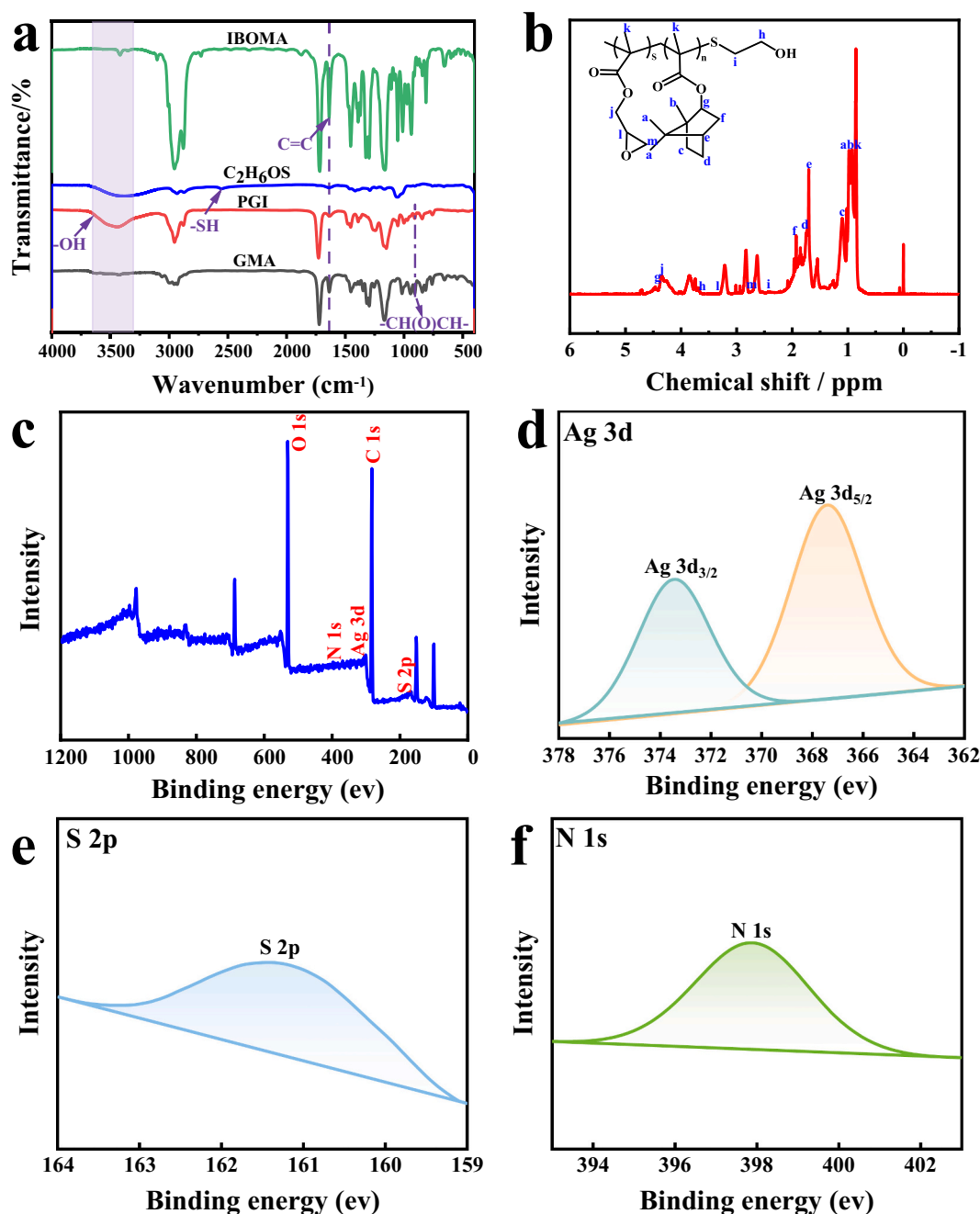


Fig. 2. (a) FTIR spectrum of IBOMA, C₂H₆OS, GMA, and PGI; (b) ¹H NMR of PGI; (c) XPS spectra of E44-Ag-PGI-20; (d) XPS spectra of Ag 3d atoms of E44-Ag-PGI-20; (e) XPS spectra of S 2p atoms of E44-Ag-PGI-20; (f) XPS spectra of N 1s atoms of E44-Ag-PGI-20.

break reached 125 % but still showed excellent flexibility. Fig. 3b was the diagram of Young's modulus and the toughness of the coating. It was obvious that the stiffness of the material decreased with the increase in PGI content. The toughness of the materials showed a positive correlation with the content of PGI. Overall, the coating prepared in this work had excellent mechanical properties and fully met the applicable coating standards.

Additionally, it can be seen from the stress-strain diagram that the stress decreases with the addition of the PGI polymerized soft chains. We believe that the introduction of PGI affects the cross-linking density of the network, which in turn affects the mechanical properties of the coating. Therefore, we calculated the cross-linking density of the coating. It can be seen from Table 1 that the crosslinking density of E44, E44-PGI-20, and E44-Ag-PGI-20 first decreases and then increases,

indicating that the introduction of PGI affects some of the cross-linking sites and thus reduces the crosslinking density of the coating. In addition, the addition of AgNPs hinders the movement of chain segments in the network, causing the *T_g* of the system to increase, thereby increasing the cross-linking density of the coating, which is also consistent with the mechanical properties test.

In addition, hardness was also one of the critical performance indexes to measure the applicable conditions of the coating. In this work, Shore hardness D was used to evaluate the hardness of the coating. Fig. 3d illustrated that the introduction of polymer PGI soft chain reduced the hardness of the coating to a certain extent, and the minimum hardness could still reach 40 HD, which was enough to guarantee that the coating avoided physical loss. At the same time, the E44-Ag-PGI-20 coating had excellent adhesion (Fig. S4), which provided the

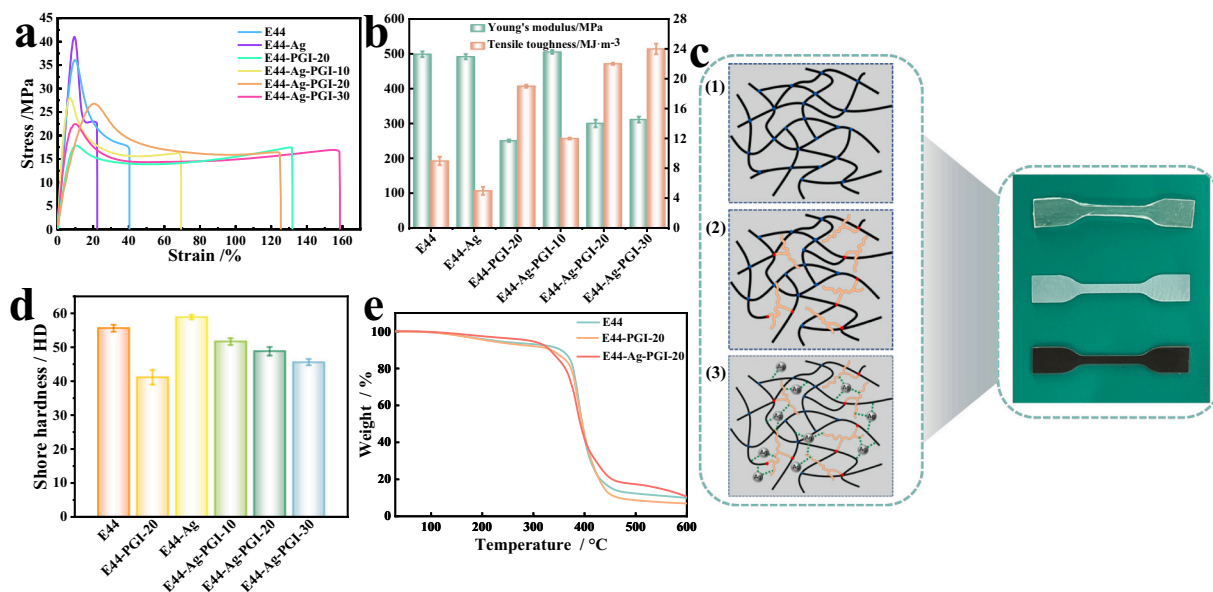


Fig. 3. (a) Stress–strain curves of E44, E44-PGI-20, and E44-Ag-PGI-x; (b) Young's modulus and tensile toughness of E44, E44-PGI-20, and E44-Ag-PGI-x; (c) The three-dimensional network structure of the coating and (d) Shore hardness of E44, E44-PGI-20, and E44-Ag-PGI-x. (e) TGA curves of E44, E44-PGI-20, and E44-Ag-PGI-20.

Table 1

Cross-linking density and calculation index of E44; E44-PGI-20; E44-Ag-PGI-20.

Sample	$T_{g\ DSC} (^{\circ}C)$	$E_{R\ Tc} \text{ (MPa)}$	$\nu \text{ (mol}\cdot\text{cm}^{-3})$
E44	33.3 ± 0.13	4.201	2.02×10^{-3}
E44-PGI-20	13.0 ± 0.23	1.379	8.78×10^{-4}
E44-Ag-PGI-20	29.1 ± 0.19	3.344	1.70×10^{-3}

prerequisite for it to be firmly fixed to the surface of subsea equipment. The mechanical properties of epoxy resin coating could be much higher than that of silicone coating, and the prepared coating had excellent mechanical properties.

3.3. Thermal stability

The thermal stability of E44, E44-PGI-20, and E44-Ag-PGI-20 coatings was evaluated by thermogravimetric analysis. The TGA curves of the coatings were depicted in Fig. 3e. The thermal decomposition of all coatings occurred between 100 and 450 °C, where there was a slight mass loss from 100 to 300 °C, mainly due to the thermal desorption of organic small molecules, and thermal desorption occurred at 300 °C. The trend of the curve from 300 to 450 °C changed abruptly. Most organic matter began to undergo oxidative decomposition, and the polymer network experienced thermal cracking, the main compound of which was bisphenol A [45]. It could be seen that with the addition of PGI and AgNPs, the maximum decomposition temperature of all coatings was almost 390 °C, indicating that the addition of PGI and AgNPs had little influence on the thermal stability of epoxy resin. Since AgNPs did not participate in thermal decomposition, there will be a residual amount of E44-Ag-PGI-20 coating. However, since AgNPs did not participate in thermal decomposition, there would be some residual amount of E44-Ag-PGI-20 coating. The DTG curves of the coatings were shown in Fig. S3. The E44-PGI-20 and E44-Ag-PGI-20 coatings have thermal decomposition near 336 °C due to the thermal decomposition of PGI polymer chains occurring in the coatings. In a word, introducing PGI and AgNPs gave the coating good thermal stability.

3.4. Morphology of the E44-Ag-PGI-20 coatings

Notably, the adhesion of fouling organisms (such as diatoms) to the coating was related to the surface morphology and roughness. Therefore, the surface morphologies of E44, E44-PGI-20, and E44-Ag-PGI-20 coatings were observed by SEM, as indicated by Fig. 4a. It could be observed that the surface of E44 and E44-PGI-20 coatings were relatively flat and smooth without protections and folds, indicating that the coating had good film-forming performance. Compared with Fig. 4a (1) and (2), it could be intuitively observed that the AgNPs were evenly distributed on the coating from the SEM image of the E44-Ag-PGI-20 coating. Furthermore, the arrangement was relatively dense, with no large agglomerated particles. At the same time, the surface roughness of the coating was increased. In addition, the surface roughness of E44-Ag-PGI-20 coating before and after 14 days of immersion in artificial seawater was evaluated digitally by AFM test. Among them, Ra and Rq values were important parameters for assessing the coating roughness, and the results were depicted in Fig. 4b. It could be seen that the values of Ra and Rq of E44-Ag-PGI-20 coating did not change significantly before and after soaking, but was still relatively flat with good smoothness. This was due to the interaction between groups in the prepared new epoxy resin antifouling system. This made the cross-linked structure compact, with excellent hydrolysis resistance, so the coating had a stable and good surface topography. Moreover, the small changes in Ra and Rq values of the coating indicated that the coating was relatively hard, which also ensured that the coating could have a long service life. In a word, the coating had good smoothness and low roughness at both macro and micro and nano scales, which had certain advantages for resisting fouling adhesion. Fig. 4c was the EDS element atlas of E44-Ag-PGI-20 coating, from which S and Ag elements could be observed to be evenly distributed on the coating, indicating that the prepared anti-fouling agent PGI had been successfully introduced into the cross-linking network of the coating through in-situ polymerization. AgNPs and coating systems were compatible, with no apparent AgNPs agglomeration phenomenon.

3.5. Wettability and surface energy tests

Wettability was a vital evaluation index to judge the hydrophobicity

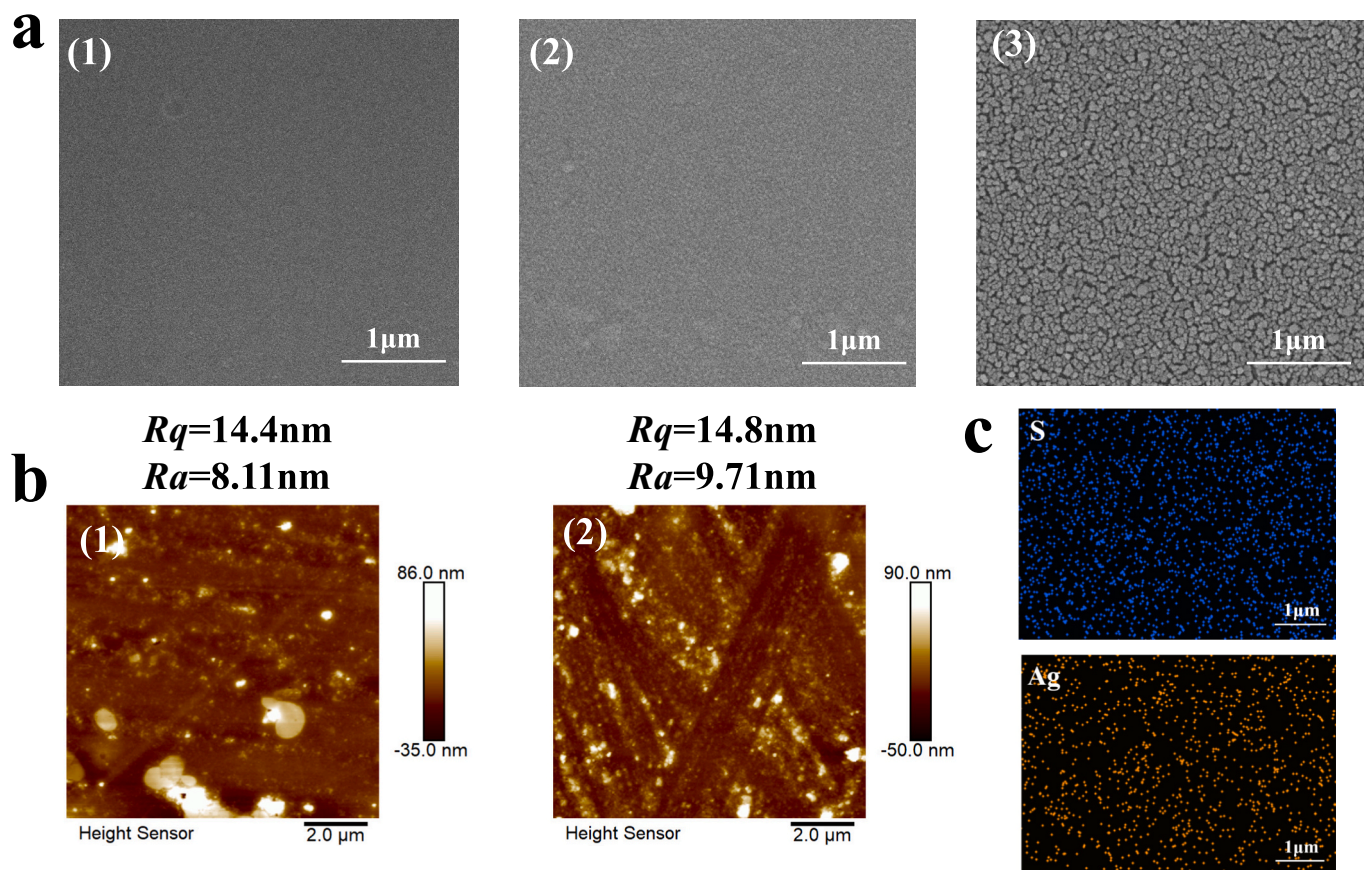


Fig. 4. (a) SEM images of the (1) E44; (2) E44-PGI-20 and (3) E44-Ag-PGI-20 coatings. (b) Atomic force microscopy (AFM) images of the E44-Ag-PGI-20 coatings. (1) before and (2) after immersion in artificial seawater for 14 days. (c) EDS images of E44-Ag-PGI-20 coating.

and adhesion resistance of coatings. As shown in Fig. 5b and c, coatings of different proportions decreased slightly after soaking in ASW. The water contact Angle (WCA) of E44 before and after soaking was about 77.13° and 75.38° , which indicated that the coating cured with D400 and E44 showed hydrophilicity. Meanwhile, the prepared PGI was in situ polymerized into E44 three-dimensional network in this study. The results showed that the coating (WCA) of E44-PGI-20 was 92.96° , which was due to the hydrophobic carbon cage structure (Fig. 5a) of the IBOMA component in PGI, which prevented the “invasion” of water molecules to the cross-linked network and changed the wettability of the coating, from hydrophilic to hydrophobic. On this basis, after adding AgNPs, wettability changed significantly compared with E44, and the WCA of E44-Ag-PGI-20 coating was 99.84° , mainly attributed to the increase of surface roughness of the coating by AgNPs. Most importantly, the E44-Ag-PGI-20 coating remained hydrophobic after soaking (WCA 96.43°), indicating a stable hydrophobic property of the coating. In addition, this property would weaken the interaction between the coating and microorganisms to play a certain anti-fouling performance [46].

As we all know, the wettability of the coating reflected the surface energy (SE) of the substrate to some extent. The change of SE also reflected the degree of molecular cohesion looseness. Hence, the Owens two-liquid method was used in this work to calculate the SE of the coating (Table 2). The SE of the coating decreased with the increase of PGI content, resulting in a decrease in intermolecular cohesion. However, the addition of AgNPs increased the SE of the coating, thus increasing its cohesion. The reinforcement effect of AgNPs was further demonstrated.

3.6. Antifouling performance

In the marine environment, bacteria could be considered as one of the critical marine biofouling because the initial attachment of planktonic bacteria to the basement membrane was reversible. However, bacterial secretions would provide a good living space for small or large fouling organisms over time. Once formed, this process would cover the surface of the material for a long time and was difficult to remove. Therefore, excellent antibacterial adhesion performance was essential in improving the antifouling coatings' effect. This study used *P. aeruginosa* and *S. aureus* to evaluate the antibacterial effect by studying the number of colonies adhered to the coating. As indicated by Fig. 6a, the number of colonies adhered by the prepared coatings to the two types of bacteria gradually decreased compared to the control group. The resistance to *P. aeruginosa* was particularly evident. At the same time, the antibacterial rate of the coating was calculated with the sample of the control coating, and the results were shown in Fig. 6b. Compared with the control group, the antibacterial efficiency of E44 against the two types of bacteria was 8.59 % and 8.69 %, respectively. Because a small number of amine groups that were not involved in the reaction had certain antibacterial properties during the curing process of the epoxy resin. The antibacterial efficiency of E44-PGI-20 containing PGI components was 58.21 % and 46.5 %, respectively. This was due to the complex bicyclic monoterpene structure of borneol, and its surface stereochemistry would affect the adhesion of bacteria. Moreover, the antifouling activity of borneol molecules would not be lost by its polymerization to long chains [37,39]. In addition, E44-Ag-PGI-20, which contained the same amount of PGI components, had an antibacterial efficiency of 97.69 % and 95.9 % against the two bacteria, respectively. This was because AgNPs can penetrate cells, inducing the formation of reactive oxygen species (ROS) in cells and then disrupting cell signal transmission. At the

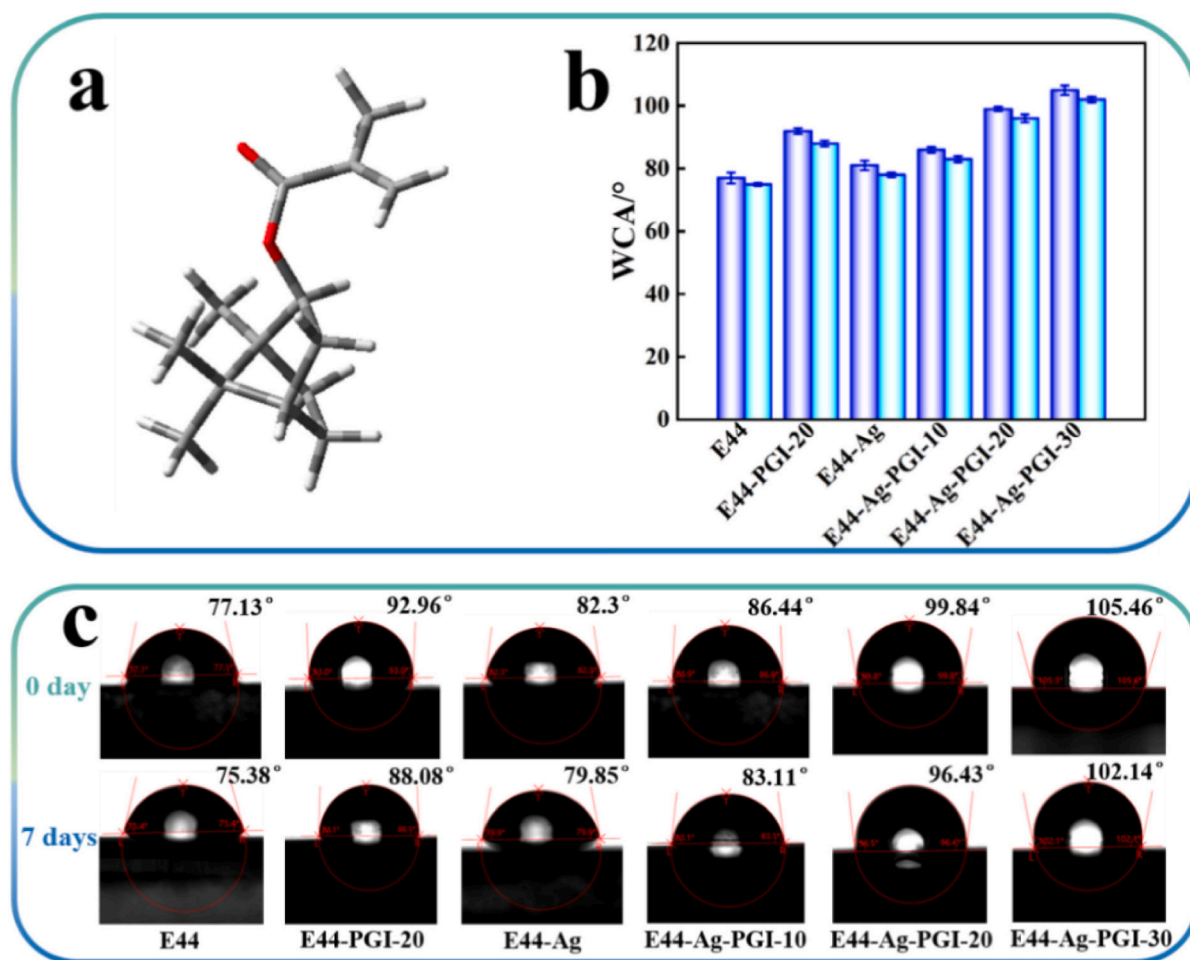


Fig. 5. (a) The stick-like molecular formula of natural borneol. (b) Static water CA and (c) CA images before and after immersion in ASW for 7 days.

Table 2

The SE of different coatings before and after immersion in ASW for 7 days. Unit ($\text{mJ}\cdot\text{m}^{-2}$).

Sample	E44	E44-PGI-20	E44-Ag	E44-Ag-PGI-10	E44-Ag-PGI-20	E44-Ag-PGI-30
Before	42.44	37.22	42.93	41.73	39.01	38.53
After	43.07	38.28	43.36	42.25	40.01	39.72

same time, the hydration layer was adsorbed, and Ag^+ was released, thus achieving a sterilization effect [47]. (See Fig. 6c for the synergistic antibacterial mechanism). The Ag^+ release rate was depicted in Fig. S5, and the dissolution rate became stable from fast, contributing to the system's stable sterilization performance. Hence, the antibacterial experiment showed that the synergistic antibacterial effect of PGI and AgNPs could give the coating an excellent antibacterial effect. Meanwhile, as shown in Fig. 6d, the antibacterial rate of gram-positive and gram-negative bacteria in the coating we prepared was much higher than in other literature [36,48–54].

In addition, anti-biofilm contamination is also an essential aspect of marine antifouling coatings that cannot be ignored. The biofilm formation was induced by prolonging the culture time of *Pseudomonas aeruginosa*, and the anti-biofilm experiment was carried out. And the optical density (OD) was proportional to the quality of the biofilm [54]. Fig. 7a illustrated that the sample of the control group had been entirely covered by biofilm, and the OD value was 1.24. The OD value of E44 was 1.17, and it slightly resisted biofilm adhesion. With the addition of PGI

and AgNPs to the coating, the OD value of E44-Ag-PGI-x gradually decreased, of which E44-Ag-PGI-10 was 0.64, E44-Ag-PGI-20 was 0.62, and E44-Ag-PGI-30 was 0.43, indicating that the biofilm attached to the coating gradually reduced. In summary, the results of the antibiofilm test were consistent with those of the antibacterial test. The synergistic effect of PGI and AgNPs could effectively improve the biofilm resistance of the coating and reduce the adhesion of biofilm.

After microorganisms (such as bacteria) form biofilms, they would provide suitable conditions for protists (such as algae and fungi) to attach to the surface of the materials, resulting in algae and other organisms to attach, causing harm to the ocean. Therefore, in this study, the adhesion of diatoms to the coating was studied by using the *Pleurosigma* sp. as the research object. As could be seen from Fig. 7b, compared with the control group ($7.15 \pm 0.98\%$), E44 coating ($4.94 \pm 0.93\%$) was not favored by diatoms because diatoms tended to adhere to the hydrophobic surface [55], while E44 coating was hydrophilic and slightly resistant to the adhesion of diatoms. After adding AgNPs and the antifouling agent PGI, the adhesion of E44-Ag-PGI-10, E44-Ag-PGI-20, and E44-Ag-PGI-30 coatings to diatoms were ($1.33 \pm 0.33\%$), ($0.8 \pm 0.2\%$) and ($0.5 \pm 0.12\%$), respectively. This was because AgNPs could affect algae growth by destroying and degrading cell walls. At the same time, the presence of IBOMA components in antifouling agent PGI reduced bacterial adhesion and inhibited biofilm formation, allowing the coating to resist diatoms. The anti-diatom experiments showed a uniform trend with the results of antibacterial and anti-biofilm experiments, indicating that the introduction of AgNPs and antifouling agent PGI could improve the antifouling ability of the coating.

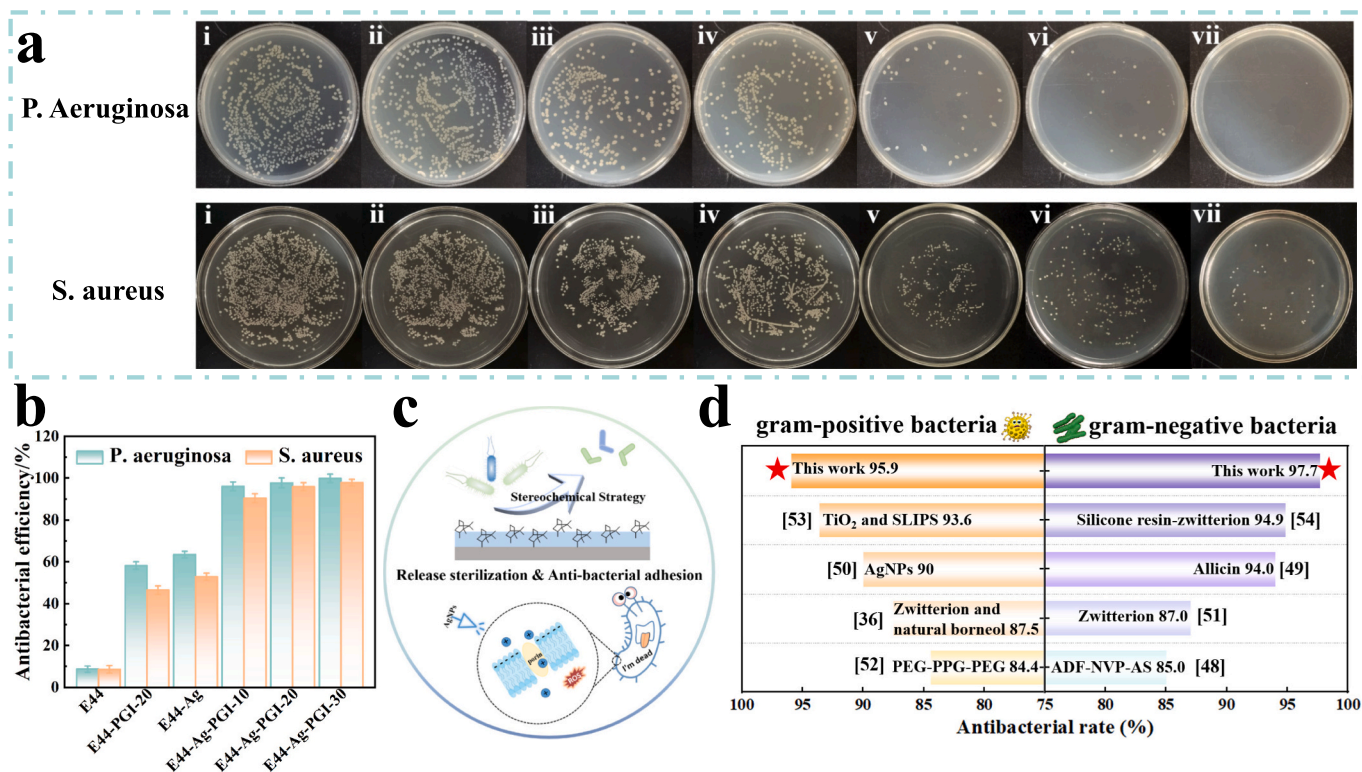


Fig. 6. (a) Digital photographs of antibacterial (*P. aeruginosa* and *S. aureus*) test of the (i) Control group, (ii) E44, (iii) E44-PGI-20, (iv) E44-Ag, (v) E44-Ag-PGI-10, (vi) E44-Ag-PGI-20, (vii) E44-Ag-PGI-30. (b) Antibacterial rates of E44, E44-PGI-20, and E44-Ag-PGI-x for *P. aeruginosa* and *S. aureus*. (c) Double strategy anti-bacterial mechanism diagram of coating. (d) Plot of the antibacterial rate of gram-positive and gram-negative bacteria comparing the E44-Ag-PGI-20 with other reported marine antifouling coatings.

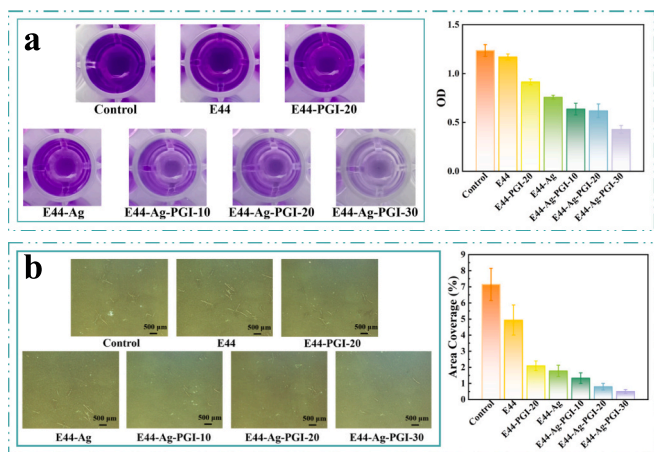


Fig. 7. (a) The images of biofilm stained with crystal violet and the optical density (OD) of crystal violet residue in the biofilm at $\lambda = 589$. (b) The images of diatom (*Pleurosigma* sp.) adhered on E44, E44-PGI-20, and E44-Ag-PGI-x and statistical chart showing *Pleurosigma* sp. density. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.7. Anti-corrosion performance

The coatings prepared by us not only had satisfactory antifouling but also had excellent anti-corrosion properties. As indicated by Fig. S6, the anti-corrosion of coatings can be seen in the Bode and Nyquist diagrams. The $Z_{[0.01]Hz}$ value of the E44-Ag-PGI-20 coating was greater than the $Z_{[0.01]Hz}$ value of the pure iron sheet. The impedance arc Angle of E44-Ag-PGI-20 coating was much greater than that of pure iron sheet. This was

because epoxy resin could be solidified and crosslinked into a three-dimensional network with good barrier properties. The addition of AgNPs might fill the channels for corrosive substances to enter the three-dimensional network and extend the path of corrosive substances, thus having an excellent anti-corrosion effect.

4. Conclusion

In summary, an environmentally friendly antifouling coating with antibacterial and bactericidal synergistic antibacterial effects was prepared based on simple and effective surface stereochemistry and release sterilization dual strategies. By designing a new copolymer PGI, the coating could effectively inhibit bacterial adhesion, and AgNPs were introduced to make the coating have a bactericidal effect. In addition, the coating had good film-forming properties and exhibited excellent adhesion properties (5B) on the steel substrate. Most importantly, the coating could effectively prevent the adhesion and accumulation of *Pseudomonas aeruginosa*, biofilm, and diatoms. This was mainly due to the combination of the anti-adhesion effect of the stereochemical structure of the borneol surface and the bactericidal effect of AgNPs, thereby improving the antifouling effect of the epoxy resin coating. Hence, the dual-strategy antibacterial proposed in this study gave some inspiration for environmentally friendly marine antifouling coatings.

Declaration of competing interest

All co-authors have seen and agree with the contents of the manuscript and there is no financial interest to report. I would like to declare on behalf of my co-authors that work described is original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the manuscript that is enclosed.

Data availability

This article describes in detail the experimental procedures and the complete characterization tests reported for all materials. Further raw/processed data required to reproduce these findings cannot be shared at this time due to technical or time limitations.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2023.107837>.

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